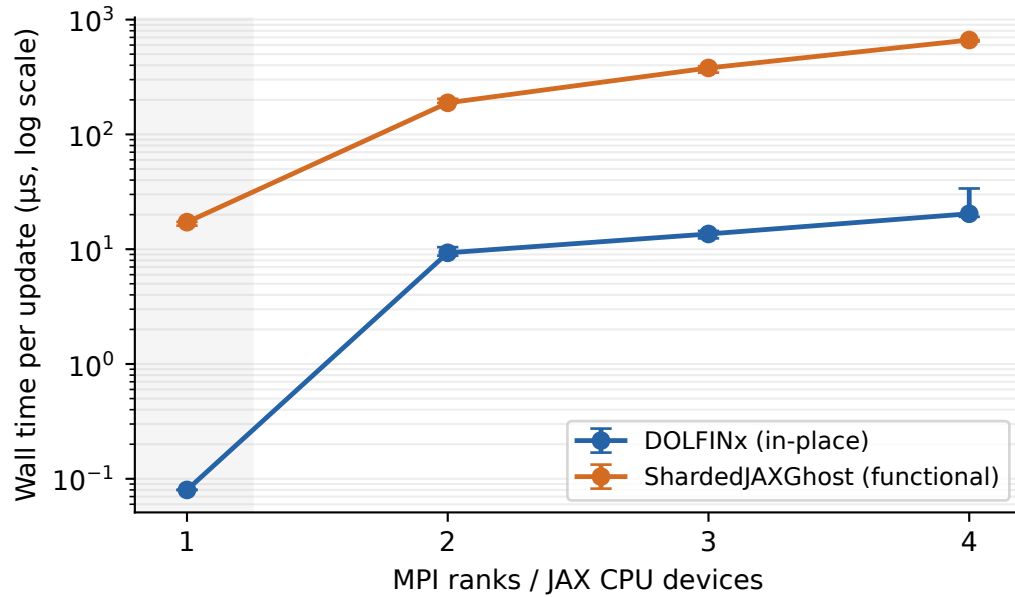


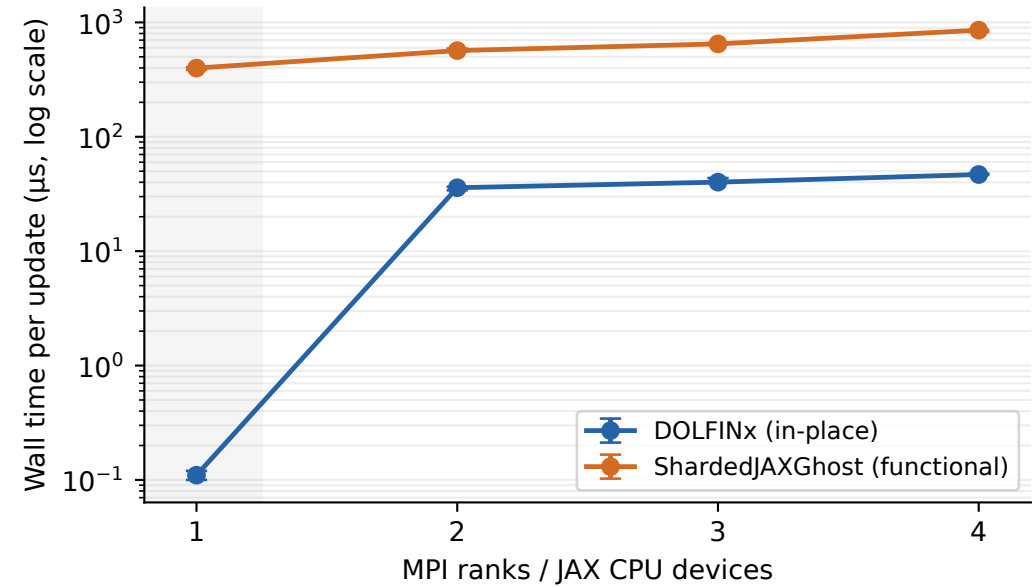
CPU forward ghost updates: fixed-workload scaling

Apple M2 Pro • float64 • one MPI rank per JAX CPU device • no CPU affinity

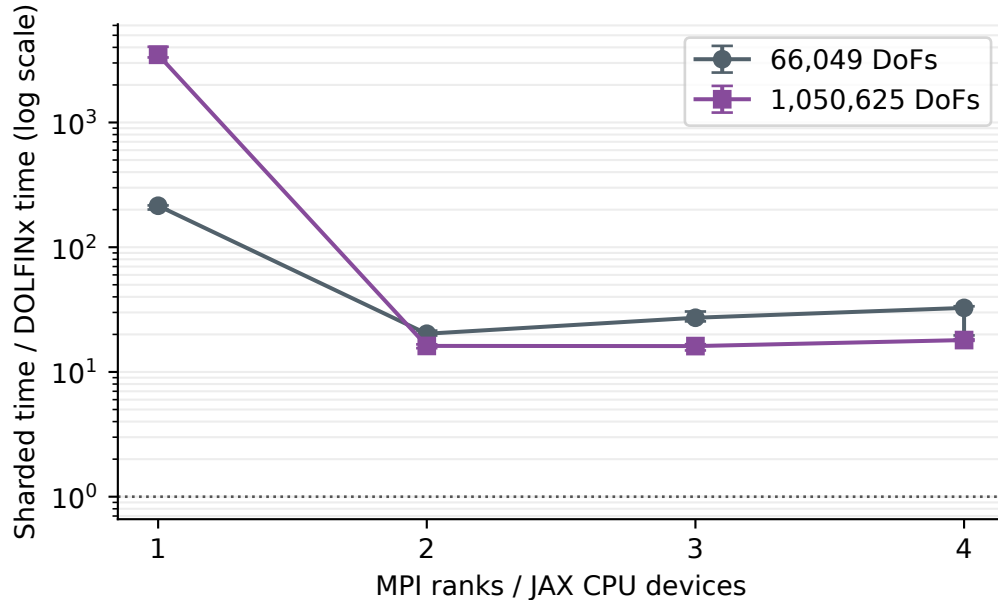
66,049 DoFs • 256 × 256 subdivisions



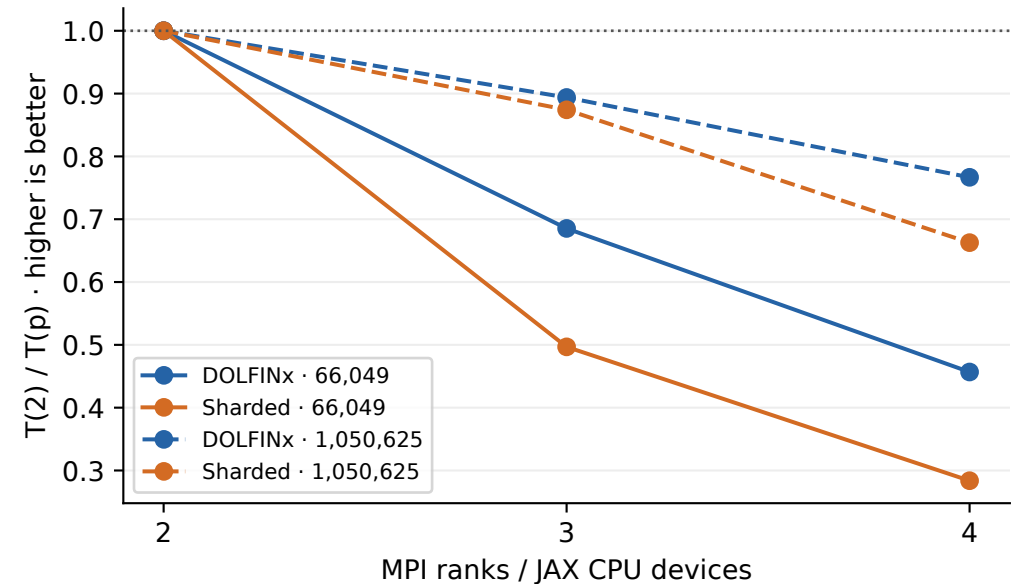
1,050,625 DoFs • 1024 × 1024 subdivisions



Backend time ratio • lower is better for JAX



Communication scaling relative to 2 ranks



Points: median of 3 independent launch medians. Bars: range of launch medians.
Each launch: 7 batches × 100 updates; sample = maximum rank time / 100. Setup and compilation excluded.
1 rank has no remote ghosts; no T(1)-based speedup or ideal 1/p line is asserted.